Certified Emulation at the Busy Beaver Frontier: A Non-Euclidean Benchmark Where Learning Meets a Provable Wall

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DeepMath 2026 One-Page Abstract

The busy beaver value $BB(n)$ is the longest finite runtime of any n-state Turing machine started on a blank tape. $BB(5)$ was settled in 2024 with a computer-checked proof. $BB(6)$ is blocked by a handful of machines whose tapes secretly run Collatz-style arithmetic. This paper studies two of these machines and surveys the rest. The approach is empirical first: the machine's event log, embedded in logarithmic, simplex, and hyperbolic coordinates, reveals dynamics piecewise affine in a five-integer state. This description is then proven correct in the Lean proof assistant, down to individual steps, with a minimal axiom footprint. The result is a machine-checked theorem: if a simple five-integer recurrence never enters an explicit finite set of bad configurations, the machine never halts. The converse pressure is also proven: entering a bad configuration forces the machine to halt. The open problem is thereby moved, without loss, from a Turing machine to a short arithmetic recurrence. The renewal map is derived in closed form with a width-five fatal window around the diagonal; computational evidence closes two mechanical routes (no congruence invariant up to modulus 384 excludes the window; exponential factor complexity, so no conjugacy to a rotation). Measuring the recurrence over its whole seed space shows the set of dying seeds has natural density zero at the Borel–Cantelli rate ($\approx 12/X$), so the studied machines' seeds are typical survivors. This density-zero is not merely measured: the Gauss-type inducing map admits a transfer-operator proof route (a.c. invariant measure [6]; a uniform spectral gap with non-integrability verified from the branch data [8, 7]; a local limit theorem [9, 10]), reducing the density form to one arithmetic exponential-sum estimate—Tao's hardest step. A two-ended harness (forward orbit against backward halt cone) classifies all 1,094 open six-state machines: zero halts; four fixed-dimension (exactly the certified ones); an eleven-machine *array class* with a validated word-rewrite calculus; and 307 single-law machines with regime structure. Certified predictive emulation (this paper) sits between kernel checking and the $\mathsf{MIP}^* = \mathsf{RE}$ verification endpoint; every empirical claim carries a horizon (10^9 steps). For the theory of learning, the pipeline is representation learning carried to a proof, and it is offered as a benchmark: unbounded, exactly labeled, law-governed sequence data on which a learned representation has a provable ground truth, with a two-prover game that verifies the forecasts of any learned emulator at constant cost per check. A from-scratch network bears this out: it learns the grammar ($C \to T$ at 0.99, TT forbidden) yet its 20-step accuracy falls to the base rate across a width/depth/context sweep while the certified emulator stays exact—the long-horizon wall is arithmetic, not capacity, exactly as the density result predicts.

The central certified statement, one per machine:

Theorem 1. *At Lean-kernel axioms* {propext, Quot.sound}*:* $\mathsf{SafeMetaOrbit} \to \neg\mathsf{RawHalts}(\mathsf{blankStart})$*;*

configurations outside the guarded region halt within ~ 50 *steps, so* SafeMetaOrbit *is the halting problem restated as arithmetic.*

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